

RESEARCH ARTICLE | AUGUST 10 1990

High throughput x-ray astronomy

Paul Gorenstein

AIP Conf. Proc. 211, 297–313 (1990)

<https://doi.org/10.1063/1.39658>



View
Online



Export
Citation

Articles You May Be Interested In

Living off the land: The practical low-cost approach to piloted Mars missions

AIP Conf. Proc. (January 1995)

High energy astrophysics from the moon

AIP Conf. Proc. (July 1990)

Russian aluminum-lithium alloys for advanced reusable spacecraft

AIP Conf. Proc. (January 1998)

HIGH THROUGHPUT X-RAY ASTRONOMY

Paul Gorenstein

Harvard-Smithsonian Center for Astrophysics

ABSTRACT

Many of the scientific objectives of x-ray astronomy for the 1990's and the twenty-first century require high throughput. Three phases of high throughput telescopes are discussed: 1m^2 , 10m^2 , and 100m^2 effective areas. The era of the 1m^2 telescope will commence in the late 1990's with the launches of XMM and LAMAR. A 10m^2 facility could be built with existing technology and launched as a free-flyer on a rocket like the Soviet Energia. With its relevance to x-ray astronomy's scientific objectives the 10m^2 facility should be given high priority as the first major initiative of the twenty-first century. A 100m^2 facility could occur in the middle of the twenty-first century. Although it is rather speculative, an appropriate site for the 100m^2 telescope is a lunar base where x-ray optics can be manufactured in situ.

I. INTRODUCTION

In the future, there will be increasing emphasis upon deriving unambiguous results directly from x-ray observations. They should be as quantitative and as definitive as those from optical telescopes. Hence, x-ray measurements must be carried out with much more precision and for many times more objects than they were on the first generation of satellites.

Table 1 is based upon various papers presented in the initial sessions of this workshop. Their common theme is a requirement for a level of throughput that is much larger than any missions currently scheduled. The throughput ranges from a few square meters for surveys to about 10 m^2 for studies of large scale structures. Several speakers described a need for systems as large as 100 m^2 to study very distant galaxies, the origins of the x-ray background, and for ultra-fast timing. A very high throughput x-ray telescope was previously recommended by the Task Group on Astronomy and Astrophysics of the Space Science Board in a study conducted by the National Academy of Sciences for the period 1995-2015 (Nat. Acad. Press., Wash. DC 1988). For nearly all of these objectives good angular resolution is needed to image, resolve sources from background, and avoid confusion. In addition, it is essential to have good spectral resolution to resolve lines from continuum. Above 2 keV, the energy resolution should be at least as good as that of a CCD, i.e. about 100 eV, for Fe line measurements. Below 2 keV the spectral resolution should be much better, $E/\Delta E > 100$, for resolving lines in the rich complexes emitted by plasmas with temperatures below 20 million degrees.

To illustrate the throughput requirements of a twenty-first century facility, consider the study of an object detected in a deep survey measurement of the *Einstein* Observatory IPC carried out in 4×10^4 seconds of net observing time. This object has a flux of 7×10^{-14} ergs/cm²-sec in the 0.8-3.5 keV band. The number of counts detected above background is about one-hundred, which is enough only for existence, position, and crude spectral information. On the basis of optical identifications, the typical object is a quasar at $z = 0.5$. Future goals would be to measure its redshift, and obtain information concerning physical conditions in the accretion disk around the central black hole, directly from x-ray observations. It is generally accepted that quasars are driven by the same mechanism responsible for x-ray emission in Seyfert galaxies. Matsuoka et al, 1989 and Nandra et al, 1989 have detected features at about 6 to 7 keV in the spectra of several Seyferts observed by *Ginga* which appear to be iron lines with equivalent widths of about 100 eV. George et al, 1989 describe a model in which the iron lines arise from the fluorescence of clouds or filaments near the central engine emitting the continuum radiation.

Table 1

**Scientific Objectives For X-ray Astronomy in the Twenty-first Century
and the Instrument Requirements**

Speaker Scientific Problem	keV	Eff. Area	Ang. Res.	Spec. Res. E/ Δ E	Time Res.	Sensitivity and Other Requirements
R. Rosner Stellar Deep Survey	0.1-5	0.3m ²	3"	10 ³		Large Sample, 10 ⁻¹⁵ ergs/cm ² s
D. Lamb X-ray Timing, cont. presence in Space	0.5-10				0.1 msec	Multiple Modest Missions
G. Fabbiano High Sensitivity Obsv. of Normal Galaxies	0.1-10	10-100m ²	1"			Large Samples Large Distance
R. Mushotzky AGN Observations	1-10	$\geq 2\text{m}^2$	$\sim 15''$	40		Large Distance 5 $\times 10^{-16}$ erg/cm ² s Need Fe Sensitivity
M. Geller Large Scale Structure	1-10	3m ²	10"			Large Cluster Sample Large area of sky
K. Wood Fast Timing Measurements	1-20	100m ²			10 ⁻⁶ sec	Post-XTE Mission Need Associated Imager
E. Boldt All Sky Background Isotropy	2-10	> 1m ²				All Sky Coverage 2 $\times 10^{-14}$ ergs/cm ² s
R. Griffiths Origin of X-ray Background	0.5-10	10m ²	$\leq 1''$	10 ³		10 ⁻¹⁹ ergs/cm ² s

According to their model the iron lines are broadened to a width of about 1 keV by a combination of doppler shifts from the high velocities of clouds in an accretion disk and gravitational effects from the black hole. The line profile reflects the velocity distribution of the clouds. In order to measure the red shift, width, and structure of an iron line it should be detected with a significance of at least 5σ . With an equivalent width of only 100 eV and an actual width of 1 keV, at least 250 counts from the line above the continuum are required. The requirement of 250 counts detected in the line (redshifted from $z = 0.5$) translates into 45,000 total counts between 0.8 and 3.5 keV or 450 times as many counts as were obtained in the *Einstein* deep survey. Since many objects should be observed, the measurement must be accomplished in 10^4 seconds or about one-quarter the time of an *Einstein* deep survey. Hence, an instrument with 1800 times more throughput than the *Einstein* Observatory is required, or 3×10^5 cm² of collecting area. Furthermore, the collecting area must be maintained to at least 5 keV to include the red shifted line. (The *Einstein* Observatory had very little area at 5 keV.)

Consider another objective described at this workshop, the study of the large scale structure of the universe up to $z = 0.5$ by mapping the distribution of clusters of galaxies in three dimensions. The CfA red shift survey has revealed a structure of surprisingly large scale (Geller and Huchra, 1989). It is based upon measuring individual galaxies and can only be carried out to $z \simeq 0.1$. Detecting clusters of galaxies through their x-ray emission and measuring their redshift from the Fe line seems to be the best (and perhaps the only) method of extending the red shift survey to larger distance. The criterion is that redshift of an Fe line from a typical cluster ($L_x = 10^{44}$ ergs/sec, $T = 5 \times 10^7 K$, $Fe/H = 0.3$ cosmic) at $z = 0.5$ be measured to a precision of 600 km/sec or 1 part in 500. The equivalent width of the line is 300 eV and its energy should be measured to ~ 10 eV. Assuming the energy resolution is 100 eV, 100 photons are required. At $z = 0.5$, the flux in an Fe line from a typical cluster is 2×10^{-7} photons/cm²-sec. Hence, with an instrument of effective area A (cm²) observing for time, T (sec) we require that:

$$AT (2 \times 10^{-7}) > 100,$$

The instrument field of view is limited to 0.5° at 5 keV by the reflectivity cut-off angle. In order to cover 1500 square degrees 3000 pointings are needed. In one year, the net time per measurement, T , (assuming 50% efficiency) is limited to 5000 sec. Therefore, $A = 10^5$ cm².

Another indicator of large scale structure is the distribution of AGN's as described by E. Boldt at this meeting. Their structure may differ from that of clusters of galaxies. A study of their distribution can be carried out simultaneously with that of the clusters. The angular resolution of the instrument should be good enough ($< 20''$) to distinguish between the point-like AGN's and the extended clusters.

II. FUTURE HIGH THROUGHPUT MISSIONS

a) Introduction

We consider three phases of future high throughput missions, characterized by successively larger instruments with effective areas of 1 m², 10 m², and 100 m² (1 keV). The phases are:

- | | |
|----------------------------------|-------------------|
| • The late 1990's | 1m ² |
| Actual Programs: | |
| XMM (0.5 m ²) | |
| LAMAR (1 m ²) | |
| • 2010 | 10m ² |
| Single Launch Into Earth Orbit | |
| • 2040 | 100m ² |
| Lunar Base | |
| In Situ Manufacturing of Optics. | |

The first phase consists of two 1 m² class missions that are scheduled for the end of the century, XMM and LAMAR. The second phase, which will start about the year 2010 is the subject of this workshop, "High Energy Astrophysics in the Twenty-First Century". The instrument has a throughput of 10 m² and is responsive to the needs of most of the scientific studies listed in table 1. A modest extrapolation from present technology indicates that it should be possible to place this facility into Earth as a free-flying observatory orbit with a single launch. The third phase of high throughput is largely speculative, with respect to its scientific justification as well as the technology needed to develop it. Based on existing technology, a 100 m² will be too large and too massive to be launched as a single entity. Given that NASA intends to concentrate its resources on establishing a lunar base, the most likely and possibly the only site for a 100 m² facility is the lunar surface. The transport of massive x-ray optics to the moon will probably prove to be more expensive than constructing the facility in situ from lunar materials.

b) Programs of the 1990's

All x-ray satellites with imaging telescopes currently scheduled or being planned during the next decade are listed in Table 2. The next significant increase in throughput will be Japan's ASTRO-D mission with four telescope modules to be launched in 1993. Its capabilities will be foreshadowed by flights of the two module BBXRT experiment aboard the Space Shuttle starting in 1990. The next important step in throughput will be the two large telescopes being built by the Danish Space Research Institute for the Soviet Union's Spectrum-X-Gamma mission. Two high throughput missions with substantially larger collecting area, improved angular resolution, and much better spectral resolution below 2 keV than prior missions are being planned for the end of the century. They are ESA's XMM and the LAMAR Space Station experiment. These two facilities will mark the first era of high throughput x-ray astronomy.

Table 2
Future X-ray Telescope Missions of Long Duration

Mission	Country or Agency	Year of Launch	Effective Area of Mirror at 1 keV	Angular Resolution*
(Einstein Observatory)	USA	1978	200 cm ²	10''
ROSAT	W. Germany	1990	430	4''
ASTRO-D	Japan	1993	1000	2.5'
SAX	Italy	1993	175	1'
SPECTRUM-X-GAMMA	USSR	1994	2500(a) 400(b)	2.5' 30''
SPEKTROSAT	Germany	1995	430	2''
AXAF	USA	1997	1000	0.5''
XMM	ESA	1998	5000	20''
LAMAR	USA	1998	10000	30''

(a) XPECT, (b) JET-X

*HPW

c) High Throughput Telescope Techniques

A twenty-first century high throughput facility (HTF) with 10 m² of collecting area can be discussed within the context of current technology. Several techniques for fabricating high throughput telescopes are listed in table 3 and discussed below.

Table 3
High Throughput Telescope Technologies

Techniques	Missions	Advantages	Comments
Epoxy Replication on CFRE Substrate	XMM-Wolter 1 Optics	Good Ang. Res.	Mandrels expensive Needs Development
Electroforming	SAX Jet-X (Spectr-X- γ) Northwestern Rocket Salyut	Good Ang. Res.	High Weight
Foil Cones	BBXRT Astro-D SODART (Spectr-X- γ)	Low Cost Light weight	Poor Ang. Res.
Bent Flats Float Glass (or Ep. Rep. CFRE)	LAMAR	No Mandrels or Simple Mandrels	K-B Optics Adaptable to Larger size

Epoxy replication

ESA's XMM Observatory has three telescope modules (Jensen, Ellwood, and Peacock, 1989). XMM's x-ray mirrors are Wolter type 1 optics with the parabola and the hyperbola integral on the same substrate. Fabrication is a two step process. Substrate shells made of carbon fiber embedded in a epoxy matrix (carbon fiber reinforced epoxy or CFRE) are laid up upon steel mandrels which are made to a shape tolerance of 10 microns and a surface roughness below 0.1 micron. A precision mandrel, slightly smaller in diameter than the CFRE substrate shells, is made of glass. Each glass mandrel is polished to an excellent surface finish and then coated with gold. Epoxy is injected into a 100 micron space between the substrate shell and the glass mandrel. The gold layer plays a dual role. It acts as a release agent for the replication process and as the reflecting material after it is transferred from the mandrel to the substrate.

The XMM method of producing mirrors requires considerable development. Although epoxy replication (upon beryllium substrates) was successful for the two small mirrors of EXOSAT (30 cm diam.), the requirements of the three large XMM mirrors (70 cm) are much greater. There are 58 nested shells in each XMM module compared to two for EXOSAT. The XMM mirrors are required to have high reflection efficiency at 7 keV. Hence, their surface smoothness must be much better than the EXOSAT mirrors which were not required to reflect efficiently above 2 keV. To date results have been encouraging (Egle et al, 1989). Each of the 58 shells requires its own highly polished mandrel. As the mandrels require a great deal of time and effort, the XMM method of mirror fabrication is expensive.

Electroforming

Two mirror systems scheduled for flight aboard satellites in the 1990's are being "electroformed", i.e. fabricated by electroplating. The first is two small modules that will be aboard SAX, an Italian national program (Butler, 1990). These mirrors constitute only a relatively small part of the SAX payload which is devoted primarily to hard x-rays. The second program is Jet-X (Wells et al, 1990), one of the two telescope experiments scheduled to be aboard the Soviet Union's *Spectrum-X-Gamma*. Jet-X will have much less throughput than the XSPECT/SODART telescopes that are also aboard *Spectrum-X-Gamma*, but is expected to have much better angular resolution.

Electroforming of x-ray telescopes is being pursued at the Czech Academy of Sciences who provided solar x-ray telescopes for the *Salyut* and *Phobos* spacecrafts, and Northwestern University (Ulmer et al, 1988) where small grazing angle telescopes optimized for short wavelength are being developed.

The SAX group has used aluminum mandrels which are machined to a double cone. The surface of the mandrel is smoothed by an acrylic dip polishing process as in the case of foil cones (see below). Similar to epoxy replication, a gold layer is evaporated over the mandrel to function both as a release agent, and as the x-ray reflecting material after it is transferred to the replica. Separation is accomplished by cooling the plated mandrel. Differences in the coefficient of thermal expansion between the aluminum mandrel and the nickel replica cause the two to separate. Performance of the SAX mirrors indicates that this process will produce mirrors with angular resolution better than an arcminute, at least in small mirrors (Citterio et al, 1988). The disadvantages of electroforming are the requirements for mandrels and high weight because of the high density of nickel. To reduce weight, compound materials consisting of thin electroformed skins with a glass sphere/epoxy filler are being developed.

Foil cones with Acrylic Lacquer

X-ray telescopes made of thin foils were first made at the Goddard Space Flight Center for the BBXRT Space Shuttle experiment (Serlemitsos, 1981) which is scheduled for flight in 1990. Aluminum foils, about 0.013 cm thick, are rolled into cones that approximate the parabolas and hyperbolas of a Wolter 1 telescope. The surface is made smooth on a scale of a few angstroms by a lacquer dip coating. The alignment and stability of the foils limit the resolution to about 2 arcminutes. Foil telescopes are low cost, relatively easy to manufacture, and light weight. They offer excellent throughput over a broad band.

Foil optics will be used in two instruments that will be launched in the mid-1990's, *ASTRO-D*, and the *XSPECT/SODART* telescope of the Soviet mission, *Spectrum-X-Gamma*. Foil mirrors offer the best ratio of weight to effective area of all mirror technologies that have been developed so far. It remains to be seen whether the angular resolution can be better than an arcminute. That would justify its use on systems that are much larger than *Spectrum-X-Gamma*.

The technique of acrylic dip polishing that has been developed to smooth the surface of aluminum foils can be used with other materials and other types of telescopes. This increases the number of options for the development of a high throughput facility for the twenty-first century,

Bent Flats, Kirkpatrick-Baez Mirrors

One telescope approach differs radically from the others. The Kirkpatrick-Baez (K-B) X-ray mirror is composed of flats that are curved slightly into parabolic cylinders. An imaging telescope consists of two successive sets of nested plates that are orthogonal to each other. Such a telescope with an angular resolution of a few arcminutes was used in a successful series of rocket flights some ten years ago. More recently, a 20 cm x 30 cm

telescope has achieved an angular resolution of 31 arcseconds half power width projected along an axis, (Fabricant, Cohen, and Gorenstein, 1988). The principal advantage of the K-B geometry is simplicity. All the reflectors are identical flats prior to bending and every reflector functions independently of every other. The problem of producing a good mirror consisted of two tasks: mass producing a thin, lightweight, stiff flat with a smooth surface, and imparting the correct parabolic curve appropriate to the plate's distance from the optic axis. The latter has been accomplished under computer control (Fabricant et al, 1986). The flat does not have to be monolithic. It can be segmented along its length and height as long as the relative alignments of the segments are maintained. A disadvantage of the K-B mirror is that for the systems commonly used the effective area above 6 keV is only half that of a Wolter 1 mirror (or its conical approximation) of the same focal length. Alternatively, a focal length that is 1.5 times larger is required to obtain equal efficiency. However, below 4 keV the K-B and Wolter mirrors are equally efficient.

Float glass is an inexpensive commercial material and has excellent surface smoothness on a scale below a micron. Its limitations are relatively high weight and deviations from flatness which will limit the angular resolution to no better than 25 arcseconds. Composite materials, for example the method of epoxy replication upon CFRE substrates being developed for XMM reflectors, offer a possible lighter weight alternative to float glass. In the K-B geometry the number of mandrels is small and they are relatively inexpensive flats. The major difficulty of composite materials is that the replication process leaves internal stresses which make it more difficult to produce high fidelity flat reflectors than cylindrical ones.

The LAMAR space station experiment is based upon K-B optics. K-B optics are potentially important for the much larger twenty-first century high throughput facility. In the K-B geometry, the mandrel are very simple, and it is possible to divide large reflectors which are nearly flat into smaller segments. Thus, the K-B geometry does allow the production of much larger mirror systems, perhaps more easily than the Wolter 1 configuration.

III. LAMAR SPACE STATION EXPERIMENT

a) Experiment Description

The Large Area Modular Array of Reflectors (LAMAR) experiment was selected as an attached payload for the Space Station. The institutions involved in the scientific development are the Smithsonian Astrophysical Observatory with overall responsibility, the Marshall Space Flight Center, the University of California (Berkeley), who have responsibility for the gratings, and the University of Chicago. The University of Arizona and the University of Texas (Arlington) will collaborate in the investigation of composites as an alternative reflector material. A sketch of LAMAR integrated into a pointing system and deployed on the Space Station is shown in figure 1. The instrument budget is very small compared to other instruments of comparable size but the cost of launch, deployment in space, and the pointing system are provided separately. Most of the observing time especially after the first year will be allocated to a NASA guest observer program.

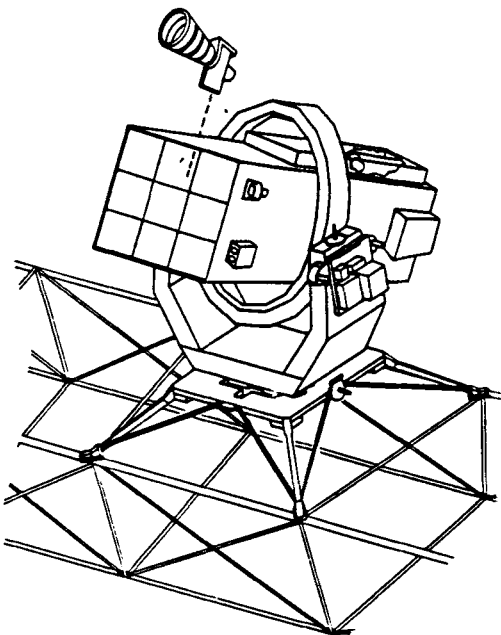


Fig. 1. LAMAR Experiment within Pointing System on Space Station.

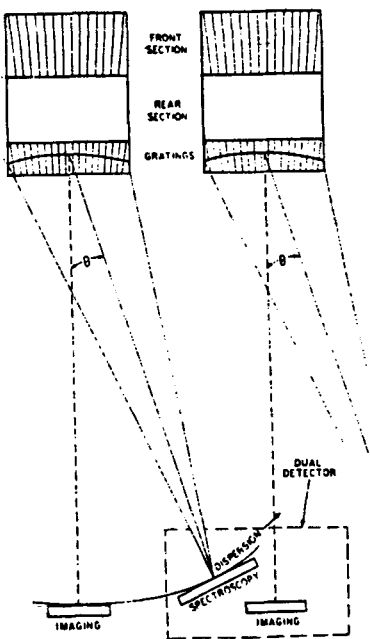


Fig. 2. Imaging and dispersive spectroscopy detectors of LAMAR.

The aperture of LAMAR is 2 m x 2 m providing a mirror effective area of about 10^4 cm² at 1 keV. The two dimensional modular array will be between 3×3 and 5×5 modules depending on how long a focal length the pointing system allows. The longer system (8m) is more desirable because fewer detectors are required. LAMAR performs imaging and dispersive spectroscopy simultaneously. Each module contains a K-B telescope, a stack of reflection gratings which intercepts about 40% of the rays, and a dual section position sensitive gas proportional counter. One section detects the direct image, the other, the dispersed spectrum which is displaced to one side. The arrangement is shown in figure 2. For two-thirds of the modules the blaze angle and line density of the gratings are optimized at 16 Å, for one-third, at 60 Å. LAMAR has no moving parts.

LAMAR's net effective area including the detector efficiency for imaging and spectroscopy are shown in table 4 and table 5. The dispersive spectroscopy is effective below 2 keV. A simulated spectral observation of Capella is shown in figure 3. A comparison of LAMAR's effective area and resolution with AXAF and XMM is shown in table 6.

b) Comparisons between LAMAR and XMM

Moderately High Resolution Spectroscopy

LAMAR and XMM are the highest throughput missions planned. They are also the only high throughput missions with angular resolution better than an arcminute and the only missions offering high throughput dispersive spectroscopy below 2 keV. The two are complementary. XMM's goal for angular resolution is 20" HPD as compared to LAMAR's goal of 30" HPW along an axis which was achieved by a smaller mirror. Although their effective area is the same at 7 keV, XMM has much better energy resolution above 2 keV thanks to CCD detectors whereas LAMAR, at least its initial configuration, has only imaging proportional counters. On the other hand, LAMAR has more effective area below 2 keV. LAMAR has large effective area in the 44-80 Å band for both imaging and dispersive spectroscopy whereas XMM does not extend longer than 30 Å. The significance of the greater spectroscopic bandwidth is illustrated in figure 3, the dispersed spectrum of Capella. LAMAR is sensitive to both bands, 8-30 Å (left) and 44-80 Å (right). XMM is not sensitive in the 44-80 Å band. Furthermore, LAMAR has three times as much effective area as XMM in the 8 - 30 Å band.

Figure 4 illustrates absorption edge features expected in many AGN's and quasars from material around the central engine (Kallman and Mushotzky, 1985). The most detectable feature is the oxygen edge where the quasar continuum flux is high. Indeed, such an oxygen edge feature was probably observed in one BL Lac object by the *Einstein* Observatory (Canizares and Kruper, 1984). The figure shows the bands of LAMAR (L-S) AND XMM (X-S) relative to the emitted spectrum for a quasar at $z = 0$, $z = 1$, and $z = 2$. At $z = 1$, the XMM acceptance band is becoming marginal whereas the red shifted oxygen absorption edge is still well within the LAMAR band. With three times as much spectroscopic effective area as XMM in the 6-30 Å band, LAMAR can study absorption features of many more AGN's.

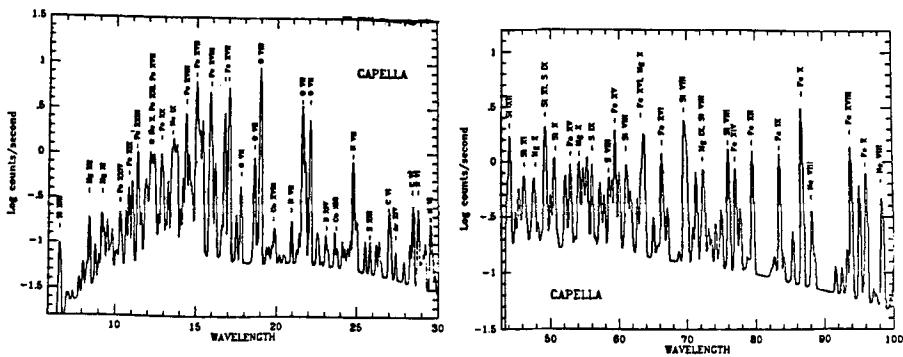


Fig. 3. Simulated spectrum of Capella in two bands, 44 - 80 Å, right panel and 8 - 30 Å, left panel as it would be observed in LAMAR. XMM is sensitive only to the shorter band.

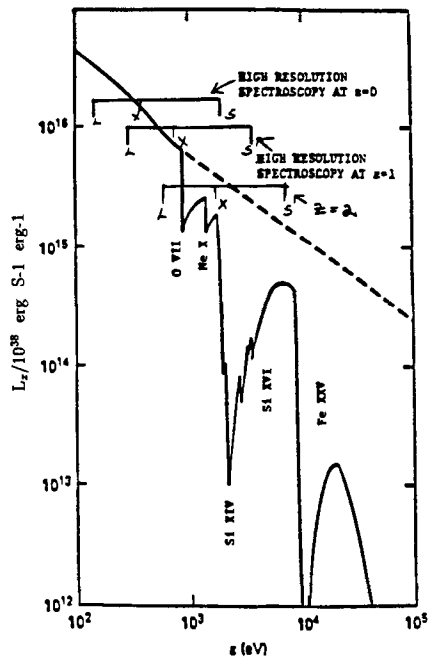


Fig. 4. Theoretical absorption line spectrum of a quasar (Kallman and Mushotsky, 1985). The lines between “L” and “S” denote the acceptance band of LAMAR in the rest frame of the quasar when it is at distances of $z = 0, 1$, and 2 . The segment of the line between “X” and “S” is the acceptance band of XMM.

Table 4
Imaging Capabilities
On Axis Ang. Resolution: 30", 50% Power width
Field of View: 40' FWHM at 2 keV, 30' FWHM at 6.2 keV

Energy (W.L.)	Net Effective Area (cm ²)*
0.28 keV (44Å)	5700
1.0 (12)	4000
4.0 (3)	2700
7.0 (1.8)	1600

Limiting sensitivity is 10^{-14} ergs/cm²-sec (0.2-8keV). It is set by confusion limit of 1 source/30 detection cells and is reached in 3×10^3 sec.

*On Axis, including detector efficiency

Table 5

High Resolution Spectroscopy Capabilities.
Field of View: 6'x 12'

Energy (W.L.)	Net Effective Area* (cm ²)	Resolution (E/ΔE)
0.15 keV (80Å)	305	450
0.28 (44)	433	260
0.50 (24)	414	320
0.78 (16)	736	230
1.0 (12)	734	180
2.0 (6)	158	80

*Including Detector Efficiency

Table 6

Comparison of Spectroscopy Capabilities at Longer Wavelength

W.L.	E	LAMAR		XMM		AXAF	
		E/ΔE	Area	E/ΔE	Area	E/ΔE	Area
16 Å	0.775keV	230	720 cm ²	300	250 cm ²	600*	60* cm ²
24 Å	0.516	315	380	400	120	850*	25*
60 Å	0.207	350	400	-	-	850	20**

*Medium Energy + High Energy Gratings, **Low Energy Gratings

Surveys

Future x-ray surveys should not only be more sensitive than ROSAT, but should also extend beyond ROSAT's 2 keV cut-off to provide spectroscopic information for distinguishing between stars and extra-galactic objects. LAMAR is much better suited for surveys than XMM because LAMAR's two-dimensional field of view is four times larger and LAMAR has a larger bandpass and larger effective area. The combination of broader field of view and higher throughput make LAMAR nearly an order of magnitude more effective than XMM for surveys. In fact, surveys can be easily carried out on the Space Station by pointing towards the zenith or during the return slews of the pointing system from the setting of a source.

Upgrade Capability

Although the Space Station is not the ideal platform for astronomy, it is unique in providing access to the instrument for upgrade. It is likely that x-ray detector technology will be considerably more advanced in a decade. We can expect imaging detectors with better energy resolution, larger format, and better spatial resolution from improved CCD's, microcalorimeters, and superconducting devices. Hence, after several years of observing, it should be possible to replace LAMAR's imaging proportional counters with future state of the art detectors. The cost is very low compared to constructing an entire new observatory with LAMAR's throughput.

IV. HIGH THROUGHPUT FACILITY (HTF)

The first goal for the twenty-first century is to develop a high throughput x-ray facility with an effective area of 10 m². It is possible to estimate its weight and volume by scaling from current technology. Table 7 is a list of the area to weight ratio for several x-ray mirror types and of the complete system.

Table 7

Effective Area to Weight Ratios of Focussing X-ray Telescopes

Telescope Type	Res.	Spacecraft	Area/Weight cm ² /kg	
			Mirror	System
Al Foil Cones	2'	ASTRO-D	30	2.5
Composites (CFRE)	20"	Spect-X-γ		
		XMM	10	2.5
Electroforming	30" - 1'	SAX, Jet-X	3-10	
Float Glass	30"	LAMAR	4	2
High Resolution	1"	AXAF	0.5	0.03

The various techniques do differ in area to weight ratio of the mirror. Foil cones are the most efficient but their angular resolution is, at least at present, not adequate for a 10 m² instrument. When the total system weight, i.e. including detectors plus structure, is considered the differences among the various techniques diminish. If we take 2.5 cm²/kg as the ratio, the mass of a 10 m² HTF is 40 metric tons. Table 8 shows the capacity and cost

of several launch vehicles. The 10 m² facility can be launched as a free flying observatory by the Soviet Union's Energia rocket. The Energia's cost launch is relatively low at \$60 M (as of 1987) and well within the cost constraints of major programs. The cost of the Energia may be artificially low because of a subsidy from the Soviet Union. This could be a Soviet contribution to an international program. Alternatively, a U.S. rocket known as the "Advanced Launch System" with capability similar to the Energia is planned for the beginning of the twenty-first century and could also be used to launch HTF.

The HTF could consist of ten modules, each with an aperture of 2 m and a focal length of 20 m. To reduce volume, the HTF can be folded up to half its length for launch as planned for both ASTRO-D and *Spectrum-X-Gamma*.

It is expected that improved dispersive and non-dispersive spectrometers will be available for the HTF. The goal is $E/\Delta E$ is 300 to 1000.

Table 8

Payload Capacity and Cost per Pound of Payload Delivered to Orbit*					
	Shuttle	Titan 4	Delta	Proton	Energia
Launch Cost Per Pound	\$6,800	\$5,100	\$3,275	\$750	\$300
Payload	24 tons	20 tons	5.5tons	20 tons	100 tons
Cost of Maximum Payload	320 M	200 M	36 M	30 M	60 M
* Newsweek Estimates August 17, 1987					

V. X-RAY ASTRONOMY FROM A LUNAR BASE

a) Introduction

At the present rate of progress the era of a 100 m² x-ray facility will occur in the middle of the twenty-first century. Its scientific justification and the methodology for its construction are both highly speculative at the present time, when the largest x-ray telescopes have been no larger than 0.1 m². An extrapolation over three orders of magnitude in throughput and a 50 year time span is likely to prove to be irrelevant. However, current scientific objectives as described by the speakers in the opening sessions of this workshop and current technology do provide some scientific justification and technical guidelines. The following scientific studies require a 100 m² facility.

- Large scale structure of the universe up to $z = 2$ through a survey in three dimensions of the distribution of clusters of galaxies and AGNs.

- Evolution of the accretion disk in AGNs up to $z = 2$ by measuring the redshift, width and structure of Fe fluorescence lines.
- X-ray observations of large samples of normal galaxies including ones at large distances.
- The x-ray background and the source content at large z .
- Very high time resolution studies of compact binaries.
- Very high resolution spectroscopy and polarization studies.

a) Construction of the X-ray Optics

Furthermore, for most of these studies the 100 m² facility should have much better angular resolution than the earlier instruments because it will be observing more distant objects of much smaller angular size. The goal is 1'' which apparently conflicts with the requirement for high throughput. By scaling current technology, the estimated mass of a 100 m² facility is 400 tons. This exceeds the launch capability of a single rocket. Currently, NASA intends to establish a base on the Moon. This goal is likely to preclude construction capability elsewhere in space. Consequently, the Moon would be the only possible site for the 100 m² facility. However, it may not be economically possible to transport 400 tons of cargo to the Moon for an x-ray observatory. The alternative is to construct the massive components, i.e. the optics, the optical bench and the pointing system in situ, from lunar materials such as O, Na, Al, Si, Ti, Fe, and Ni. X-ray detectors, star sensors and electronics which account for much less mass and are too complex to manufacture on the Moon, would be transported from the Earth. A possible configuration for the 100 m² facility is an array of 20 modules each consisting of a telescope of 5m diameter and a focal length of 50 m. Each module could have its own pointing system. If large optics are constructed on the Moon, their weight is no longer the predominant constraint that it was for the free-flying satellite. Hence, there may be no need for exotic materials. In some respects, the task of constructing optics is less difficult on the Moon than on the Earth because of the lower gravity, the vacuum, and not having to contend with vibration and acceleration during launch.

The type of optics that seems to be best suited for lunar manufacture is the bent flats or K-B design used by the LAMAR Space Station experiment. The number of mandrels that need to be sent from the Earth is minimal. Large reflectors can be segmented into smaller units. The precise figure can be tuned with cosmic x-ray sources directly on the optical bench/pointing system using the actual detectors while the star trackers keep the system in a fixed celestial position.

The process for manufacturing flats may be one of the following three:

- Glass factory: Fabrication facilities may exist because glass would be made from lunar materials for solar collectors.
- Chemical vapor deposition in the lunar vacuum,
- Electroforming using nickel or iron extracted from lunar soil: Electroforming can be used to make the structural components of the mirror, the optical bench, as well as the reflector flats.

A tele-robotic capability is needed to extract nickel, electroform the components, and assemble them.

c) High Angular High Resolution

There remains the problem of improving the angular resolution from $20''$ to $1''$ by the time a 100 m^2 facility is developed. This collecting area is three orders of magnitude larger than AXAF's. Perhaps, this goal will be feasible in fifty years as manufacturing precision improves. An alternative method is to combine focussing with a two-grid rotating modulation collimator (RMC). The RMC produces a temporal modulation at each point in the focal plane that can be transformed into finer scale angular information. By this technique, the RMCs of SAS-3, Ariel 5 and Hakucho were able to refine the angular resolution of collimated proportional counters from 15 degrees to a few arcminutes, or more than two orders of magnitude. The task of improving the resolution from $20''$ to $1''$ is actually more modest than what was achieved by these instruments. A two grid RMC with a separation of 30 m and a wire spacing of 75 microns would provide a resolution of $1''$. Diffraction would not be the limiting factor for wavelengths shorter than $5 \text{ \AA}$. The disadvantage of the RMC are a factor of four loss in effective area, additional size and complexity, and the cost of transporting some number of 100 kg pairs of wire grids to the Moon. On the other hand, the RMC improves the resolution over the entire 1 degree field of the telescope, whereas the $1''$ field of a pure focussing telescope is only about $15'$. The larger field is important for surveys. The problem of distinguishing intrinsic temporal variations from those caused by the rotation of the collimator can be solved, as it was on Hakucho, with RMCs on different modules rotating synchronously in opposite directions. Hence, the summed count rates remain constant.

VI. REFERENCES

- Butler, R. C., 1990, in *High Resolution X-ray Spectroscopy of Cosmic Plasmas*, P. Gorenstein and M. Zombeck eds. Cambridge University Press, Cambridge, UK (in press).
- Canizares, C. R., and Kruper, J., 1984, *Ap.J. (Letters)*, **278**, L99.
- Citterio, O. *et al*, 1988, *Applied Optics*, **27**, 1456.
- Egle, W., Bulla, H., and Kaufman, P., 1989, *Proc. of the S.P.I.E.*, **1160**, 432.
- Fabricant, D. G., Conroy, M., Cohen, L. M., and Gorenstein, P., 1986, in *Proc. of the S.P.I.E.*, **640**, 164.
- Fabricant D. G., Cohen, L. M., and Gorenstein, P., 1988, *Applied Optics*, **27**, 1456.
- Geller, M., and Huchra, J., 1989 *Science* **246**, 897.
- George, I.M., Nandra, K., and Fabian A. C., 1990, *M.N.R.A.S.*, (in press).
- Jensen, P. L., Ellwood, J. M., and Peacock, A., 1989, *Proc. of the S.P.I.E.*, **1160**, 525.
- Kallman, T. R., and Mushotsky, R., 1985, *Ap.J.*, **292**, 49.
- Matsuoka, M., Yamauchi, L., Piro, and and Murakami, T., 1989, *Ap.J.*, (in press).
- Nandra, K. A., Pounds, K. A., Fabian A. C., and Rees, M. J., 1989, *M.N.R.A.S.*, **236**, 39.
- Serlemitsos, P., 1981, in *X-ray Astronomy in the 1980s* NASA Tech. Mem 83848, 441.
- Ulmer, M., Matsui, Y., Bedford, D. K., Simneff, G. M., and Takar, S., 1987, *Applied Optics*, **26**, 3852.
- Wells, A., Lumb, D. H., Pounds, K. A., Stewart, G. C., Ashenbach, B., Brauniger, H., Hasinger, G., Trümper, J., Citterio, O., Scarsi, L., Peacock, A., and Taylor, B., 1990, in *High Resolution X-ray Spectroscopy of Cosmic Plasmas*, P. Gorenstein and M. Zombeck eds. Cambridge University Press, Cambridge, UK, (in press).

DISCUSSION

Claude Canizares

To me, one of LAMAR'S greatest strengths is its ability to do surveys. Have you considered spending half the observing time over several years to do a survey of stars, clusters, and AGNs?

Gorenstein

We certainly consider surveys to be one of LAMAR's important objectives. There are several options. One is performing surveys during the return slew between the rising and setting of a source. Another is to schedule a dedicated survey for a significant period of time as ROSAT plans for its first six months. This can be done by pointing the instrument in an anti-Earth direction and changing its inclination periodically. For better or worse, there is plenty of time to develop the optimum strategy.

Kent Wood

For reference, XLA at 100 m² weighs 25 tons or less with some allowance for improvement. What is the largest aperture that can fit into a Shuttle flight and on Energia?

Gorenstein

The LDEF which is due to be retrieved by Space Shuttle in January 1990 has an area of 14' × 30'. That aperture is the largest one I know of that can fit into the Space Shuttle. At 100 tons the weight-lifting capacity of the Soviet Energia is four times that of the Space Shuttle. I do not know if this translates into four times the aperture, because I am not familiar with the limitations on length and diameter of the payload.

Richard Mushotzky

Could NASA consider an "unmanned" modular spacecraft, e.g., use "lego" technology, to assemble modules spacecraft?

Gorenstein

This certainly would be important technology for scientific satellites and very appropriate for a modular high throughput facility. However, the current emphasis on establishing a Lunar base may not leave much room in the NASA program for developing a construction capability elsewhere in space.

Steven Kahn

Although you emphasized the point several times in your talk, I really want to comment again on the need for a facility with an array of telescopes that allows frequent access for instrument switch-out. In principle, if such a process could be made cost effective, it could solve a sociological problem now facing the field namely that it has become very difficult for a university-based research group to become involved in space experiments without involving a major national laboratory or industrial contractor. If LAMAR exists on the Station or the Moon, and if there are frequent flights, it might eventually become possible for university groups to build a science-specific experiment in its own laboratory, at reasonably low cost.

Gorenstein

What you suggest would be excellent for X-ray astronomy. However, at this time the prospects for frequent visits to an experiment on the Space Station (or a Lunar base) seem remote.